

	through time into space.
7(a)(i)1.	Displacement is defined as the linear distance moved by an object in a specified direction from a defined reference point.
7(a)(i)2.	Amplitude is the magnitude of the maximum value of the displacement.
7(a)(ii)	Simple harmonic motion is an oscillatory motion whereby the acceleration of an object is directly proportional to the displacement of the object from its equilibrium position, and the acceleration is always directed towards that position.
7(b)	The restoring force for the simple pendulum is the component of the bob's weight parallel to its displacement. The restoring force for the floating block is the resultant force due to the upthrust on the block by the water and the block's weight.
7(c)	At the top surface of the strip, the distances between the atoms are further away and hence it is being stretched (extension). At the bottom of the strip, the atoms are closer and hence it is being compressed.
7(d)	Since the acceleration a is directly proportional to the displacement x , where C , E , L^3 and M are constant, and it is always pointing in the opposite direction to x , the end of the strip is thus undergoing simple harmonic motion.
7(e)(i)	From Fig. 7.4, $T = 0.21\text{s}$. Since $\omega = \frac{2\pi}{T}$, $\omega = \frac{2\pi}{0.21}$

	$\omega \approx 29.9 \text{ rad s}^{-1}$
7(e)(ii)	By comparing the expression in (d) with that of the simple harmonic equation, we have $\frac{CE}{L^3M} = \omega^2$ $\frac{C \times 2.0 \times 10^{11}}{0.80^3 \times 0.150} = 29.9^2$ $C \approx 3.44 \times 10^{-10}$
7(f)	Since both the steel and the aluminium strips have the same frequency and the values of C and L, $\frac{E}{M} = \frac{E'}{M'}$ $\frac{2.0 \times 10^{11}}{0.150} = \frac{7.1 \times 10^{10}}{M'}$ $M' \approx 0.0533 \text{ kg}$
8(a)	A photon is a quantum of electromagnetic radiation with energy $E = hf$, where h